GENERIC EXACT RECONSTRUCTION OF REFLECTED-CAP DATA FROM AN ERASED FLAT BOUNDARY

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Abstract. Let a convex three-polytope in a hyperplane of $\mathbb{R}^4$ carry one reflected pyramidal cap on each facet, with the caps independently rotated about their facets. We study the proper-zero regime: at least one cap remains nonflat and at least one cap returns to the core hyperplane. The flat caps then merge with the core and erase their internal source-layer boundaries.

We define an intrinsic boundary-germ skeleton of the merged flat component. Its vertices satisfying a perpendicular-bisector neighbour test are exactly the hidden reflected apexes, provided no core vertex satisfies the same test. This yields exact reconstruction and uniqueness of every compatible decomposition. For each fixed realized core and visible-facet mask, centres violating the hypothesis lie in a finite union of proper spheres and quadrics, and hence form a measure-zero set.

Beyond the generic statement we prove two unconditional structure results. Uniqueness holds whenever every core vertex that passes the test lies on a visible hinge facet. Moreover, any two distinct compatible decompositions must differ in at least five reflected sites on each side; consequently a proper-zero datum with at most four hidden facets is unique without any genericity hypothesis. The five-site bound rests on a nonexistence theorem for an eight-point spherical configuration, proved by four polynomial certificate identities that are re-verified by exact expansion, together with an elementary positivity analysis.

We also give an exact rational reference algorithm, a certifying variant whose two comparison conditions are each provably necessary, and a 16,744-instance finite replay: reconstruction on every instance, exact certification on a deterministic 340-instance sample. The finite replay supports the implementation; it is not used as the proof.

1. Introduction

Inverse Voronoi and Dirichlet problems usually begin with a supplied tessellation, cell complex, embedded diagram, or collection of cell boundaries. The input considered here is coarser. Several reflected sites and their facet boundaries have been erased by a coplanar union before the inverse problem begins.

The model is elementary. A convex polytope P lies in a three-dimensional hyperplane $H \subset \mathbb{R}^4$, an interior point o is fixed, and the reflection of o in

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each facet plane is used as the apex of a pyramid. Each pyramid is then rotated through an independent angle about its base facet. When an angle is zero, the corresponding pyramid is coplanar with P. All such flat pyramids merge with P into one three-dimensional subset $M_J \subset H$. Only the raw union is observed; cell labels and internal flat seams are not.

The central issue is therefore not the forward reflection construction. That construction is classical and appears explicitly in Ash and Bolker's Theorem 2 [3]. The issue is whether the hidden reflected sites can be identified from the exposed boundary after the internal seams have disappeared.

A naive graph is insufficient. Taking the polygonal one-skeleton after every coplanar merge can swallow a collinear point at which the local boundary germ changes. Retaining the sides of the original pyramids repairs that example but illegitimately remembers the unknown decomposition. We instead use the noncoplanar crease graph of the maximal exposed planar sheet germs. It keeps a collinear point exactly when the local incident-sheet germ changes, and removes it when it is only a presentation subdivision.

Our main result is conditional but size-uniform.

Theorem 1.1 (informal)**.** *In dimension three, suppose a reflected-cap datum is proper-zero and no core vertex passes the intrinsic bisector-neighbour test. Then the passing vertices are precisely the hidden reflected apexes, the core is recovered by their bisector halfspaces together with the visible hinge halfspaces, and the datum is unique up to ambient Euclidean congruence. For a fixed realized core and visible mask, the excluded centres form a measure-zero subset of the core.*

At excluded centres the conditional theorem is silent, but the ambiguity it leaves open is strongly constrained.

Theorem 1.2 (informal)**.** *Uniqueness already holds whenever every core vertex that passes the test lies on at least one visible hinge facet. In general, two distinct compatible decompositions of one proper-zero raw union must differ in at least five reflected sites on each side. Hence a proper-zero datum with at most four hidden facets is unique, with no hypothesis on its centre.*

The five-site bound reduces, through an exchange graph and an elementary incidence count, to the nonexistence of a specific eight-point spherical configuration. That nonexistence is proved by a computer-assisted but fully verifiable argument: four explicit polynomial identities, found by a Gröbner search and re-verified independently by exact expansion, force the configuration into a one-parameter family that an elementary positivity analysis then empties. Section 7 contains the complete argument.

The theorems do not assert uniqueness for all data, a result in higher dimension, or novelty or priority relative to all possible prior art. Those boundaries are repeated in Section 10.

2. The reflected-cap model and exact contacts

Let $H \cong \mathbb{R}^3$ be a fixed hyperplane in $\mathbb{R}^4$, and let e be a unit normal to H. Let $P \subset H$ be a full-dimensional convex polytope and $o \in \operatorname{int} P$. Translate o to the origin and write the irredundant unit-normal description

$$P = \{x \in H : n_i \cdot x \le h_i,\ i = 1, \dots, m\}, \qquad \|n_i\| = 1, \quad h_i > 0,$$

with facets $F_i = P \cap \{n_i \cdot x = h_i\}$. For $\theta_i \in (-\pi, \pi)$ set

$$a_i(\theta_i) = h_i(1 + \cos\theta_i)n_i + h_i \sin\theta_i e, \qquad C_i(\theta_i) = \operatorname{conv}(F_i \cup \{a_i(\theta_i)\}).$$

The excluded value π collapses the apex to o. At $\theta_i = 0$, $a_i = 2h_i n_i$, the reflection of o in aff F_i: at zero angle the sites are exactly those of Ash and Bolker's forward construction [3].

Two data are called congruent when an ambient Euclidean isometry carries the core, centre, and unordered cap family of one to those of the other. Thus the angle coordinates are used only to parametrise the geometric caps.

The datum is $D = (P, o, \theta)$ and its raw union is

$$X(D) = P \cup \bigcup_{i=1}^{m} C_i(\theta_i) \subset \mathbb{R}^4,$$

retained as a bare Euclidean subset. Put

$$J = \{i : \theta_i = 0\}, \qquad I = [m] \setminus J.$$

The datum is *proper-zero* when I and J are both nonempty. Its merged flat component is

$$M_J = P \cup \bigcup_{j \in J} C_j(0) \subset H.$$

Proposition 2.1 (exact contacts)**.** *For every angle vector $\theta \in (-\pi, \pi)^m$,*

$$P \cap C_i(\theta_i) = F_i, \qquad C_i(\theta_i) \cap C_j(\theta_j) = F_i \cap F_j \quad (i \ne j).$$

Thus the cells form an embedded face-to-face polyhedral complex whose contact poset is independent of the angles.

Proof. Use the Voronoi diagram of the sites $0, a_1(\theta_1), \dots, a_m(\theta_m)$. The sites are distinct: equality of two nonzero H-components would force two irredundant outward facet normals to agree. For $x \in P$,

$$\begin{aligned} \|x - a_i\|^2 - \|x\|^2 &= \|a_i\|^2 - 2x \cdot a_i \\ &= 2h_i(1 + \cos\theta_i)(h_i - n_i \cdot x) \ge 0. \end{aligned}$$

The first factor is positive on $(-\pi, \pi)$. Hence P lies in the Voronoi cell V_0 of 0, and $F_i \subset V_0 \cap V_i$. The apex a_i is in the interior of V_i. Convexity of V_i gives $C_i \setminus F_i \subset \operatorname{int} V_i$. Therefore no point of $P \cap C_i$ can lie outside F_i.

If a point of $C_i \cap C_j$ lay outside one of the two hinge facets, the same argument would place it in one open Voronoi cell and simultaneously in a different open cell or in V_0. This is impossible. The reverse inclusion $F_i \cap F_j \subset C_i \cap C_j$ is immediate. $\square$

3. Intrinsic recovery in the proper-zero regime

For a three-dimensional polyhedral subset $Y \subset \mathbb{R}^4$, let $\mathrm{FlatReg}_3(Y)$ be the set of points having a neighbourhood in Y that is relatively open in one affine three-plane. This set is determined by Y alone.

Proposition 3.1 (proper-zero intrinsic recovery)**.** *If $I \neq \varnothing$, the raw subset $X(D)$ intrinsically determines M_J, every nonflat cap and its hinge facet, the centre o, and the fully reflected site associated with every nonflat hinge.*

Proof. The closures of the connected components of $\mathrm{FlatReg}_3(X)$ are M_J and the nonflat caps. Indeed, the zero-angle cells join the core through facets inside H and form one three-dimensional flat component, while every nonflat cap has a different affine hull. Proposition 2.1 accounts for every contact. The codimension-one adjacency graph of these closures is a star.

If $|I| \geq 2$, the unique star centre is M_J. If $|I| = 1$, let i be the nonflat index, let $V = \mathrm{vol}_3(P)$, and let $v_k = \mathrm{vol}_3 \operatorname{conv}(\{o\} \cup F_k)$. The radial pyramids tile P, so $\sum_k v_k = V$. Rotation preserves the intrinsic volume of a cap, and hence
$$\mathrm{vol}_3(M_J) = 2V - v_i > v_i = \mathrm{vol}_3(C_i).$$
Thus in the two-component case the larger component is M_J.

Fix one recovered nonflat cap, with apex A_i and hinge F_i. Let p_i be the orthogonal projection of A_i to $\operatorname{aff} F_i$, let $r_i = \operatorname{dist}(A_i, \operatorname{aff} F_i)$, and let ν_i be either unit normal to $\operatorname{aff} F_i$ inside H. Rotation about $\operatorname{aff} F_i$ preserves the radius in its two-dimensional normal plane, so
$$\{p_i - r_i\nu_i, p_i + r_i\nu_i\} = \{o, a_i(0)\}.$$
The centre belongs to the interior of M_J. The other point does not belong to M_J: apply Proposition 2.1 to the angle vector in which cap i is also set to zero. Its apex meets neither the core nor another cap away from prescribed lower-dimensional contacts. Consequently o is the unique one of the two candidates in $\operatorname{int} M_J$. Reflection of o in the recovered hinge then gives the fully reflected site. ☐

Every compatible decomposition of a given proper-zero raw union therefore has the same M_J, centre, nonflat caps, and visible hinge facets.

4. The intrinsic boundary-germ skeleton

Let an *exposed planar sheet* of ∂M_J be the closure of a maximal connected planar piece of its relative-regular locus. Internal coplanar subdivisions are not retained.

Definition 4.1 (intrinsic germ skeleton)**.** Atomise the boundaries of all exposed sheets at every endpoint contributed by any other sheet. Retain the open segments carried by at least two distinct supporting planes. Suppress a collinear degree-two point only if its complete local incident-sheet germ is unchanged through that point. The resulting canonical graph is $\Gamma(M_J)$. A vertex a *passes* if every neighbour of a in $\Gamma(M_J)$ belongs to the perpendicular bisector

$$B(o,a) = \{x \in H : \|x - o\| = \|x - a\|\}.$$

The definition is intrinsic: exposed sheet subsets, their supporting planes, and their local germs are properties of M_J as a point set. In particular, the graph stores neither a hidden cap label nor an erased internal seam.

A compatible decomposition of M_J consists of a convex core P' and hidden facets G_a indexed by sites a, where the facet set of P' is exactly the already recovered visible hinge facets together with the pairwise distinct facets G_a, each site a is the reflection of the recovered centre o in $\operatorname{aff} G_a$, and

$$M_J = P' \cup \bigcup_{a \in A} \operatorname{conv}(G_a \cup \{a\})$$

with pairwise disjoint cell interiors. The original datum is one such decomposition. For an edge f of G_a, write $W(a,f) = \operatorname{conv}(\{a\} \cup f)$ for the corresponding wall.

Lemma 4.2 (boundary inventory)**.** *For every compatible decomposition,*

$$\partial M_J = \bigcup_{i \in I} F_i \ \cup \ \bigcup_{a \in A} \ \bigcup_{f \in E(G_a)} W(a,f).$$

Proof. The boundary of P' is the union of its visible facets and hidden facets. Each hidden facet is shared with its cap and becomes internal to M_J. The remaining cap faces are the walls. Proposition 2.1 says that a wall meets another cell only in a lower-dimensional face. □

Lemma 4.3 (decomposition computation of the intrinsic graph)**.** *Starting from the inventory in Lemma 4.2, erase a side across which the two incident boundary patches are coplanar, atomise at every other sheet-boundary endpoint, and suppress only unchanged collinear degree-two germs. The result is $\Gamma(M_J)$, independently of the compatible decomposition used for the computation.*

Proof. At a relative-interior point of a patch side, exact face-to-face contact gives no transverse crossing in a patch interior. If the two incident patch planes agree, their union is locally planar and the side is absent from the crease locus. If the planes differ, the side is an intrinsic noncoplanar crease. Passing to maximal planar sheets deletes exactly the first kind. Atomisation restores every endpoint where another sheet germ begins, ends, or changes, including a collinear endpoint omitted by a polygon hull. The final suppression removes only a presentation subdivision at which the complete local germ is unchanged. These conditions depend only on ∂M_J. □

Remark 4.4 (the swallowed-corner control)**.** There are legitimate data for which two coplanar merged triangles have a collinear boundary point that a polygonal hull would drop. At that point a noncoplanar incident sheet changes, so Definition 4.1 retains it. The exact control used in the artifact is

$$P = \{x \geq 0,\ z \geq 0,\ x + y \geq 2,\ y \leq 6,\ x + z \leq 4\}, \qquad o = (1, 2, 1),$$

with the hidden facets $x = 0$ and $z = 0$. Their shared edge starts at the foot of the perpendicular from o, making the two apexes and that endpoint collinear.

5. The apex signature and uniqueness

Lemma 5.1 (only its own walls contain an apex)**.** *For $a \in A$, the boundary patches containing a are exactly the walls of the cap with apex a.*

Proof. The reflection a lies strictly outside P'. For $b \neq a$, Proposition 2.1 gives $C_a \cap C_b = G_a \cap G_b \subset P'$, so a belongs to no wall of C_b. Lemma 4.2 leaves no other boundary patch. □

Lemma 5.2 (candidate types)**.** *Every vertex of $\Gamma(M_J)$ is either a hidden apex or a vertex of the compatible core.*

Proof. By Lemma 4.3, a graph vertex is a branch point or a retained endpoint of the patch-side arrangement. Exact face-to-face contacts create no transverse crossing in a patch interior, so such a point is a patch corner. Visible-facet corners are core vertices, and wall corners are one apex and two core vertices. □

Lemma 5.3 (apexes pass)**.** *Every hidden apex passes in $\Gamma(M_J)$.*

Proof. Let a be the apex over a hidden facet G_a. For each vertex v of G_a, the two walls over the two facet edges incident to v meet along $[a, v]$. Their planes are distinct: if they were coplanar, the apex and three consecutive vertices of the convex facet polygon would be coplanar, forcing $a \in \operatorname{aff} G_a$, contrary to $o \in \operatorname{int} P'$. Thus the relative interior of $[a, v]$ is an intrinsic crease. The apex is a branch point, and v is retained because the two wall-sheet germs containing a end or change there. Conversely, Lemma 5.1 shows that every crease incident to a is of this form. Hence

$$N_{\Gamma(M_J)}(a) = \operatorname{vert} G_a \subset \operatorname{aff} G_a = B(o, a).$$

□

For a facet f of a compatible core, write $v_f = \operatorname{vol}_3 \operatorname{conv}(\{o\} \cup f)$.

Lemma 5.4 (volume identity)**.** *Let $W = \sum_{i \in I} v_{F_i}$. Every compatible decomposition satisfies*

$$\operatorname{vol}_3(M_J) = W + 2 \sum_{a \in A} v_{G_a}.$$

Consequently all compatible cores have the same volume.

Proof. The radial pyramids from o to the facets tile a convex core, so $\operatorname{vol}_3(P') = W + \sum_a v_{G_a}$. The cap over G_a is a reflection of its radial pyramid and has the same volume. Exact contacts give pairwise disjoint cell interiors, proving the displayed identity. Both M_J and W are intrinsic, so $\sum_a v_{G_a}$ and then the core volume are common to all decompositions. $\square$

Lemma 5.5 (nested site sets). *If (P', A) and (P'', A'') are compatible and $A'' \subseteq A$, then the two decompositions coincide.*

Proof. The hidden facet associated with a has halfspace $\{x : \|x - o\| \leq \|x - a\|\}$, the Dirichlet-cell description of the core against its reflected sites [3]. Therefore

$$P' = \bigcap_{i \in I} H_i^- \cap \bigcap_{a \in A} \{x : \|x - o\| \leq \|x - a\|\},$$

and similarly for P''. The inclusion $A'' \subseteq A$ gives $P' \subseteq P''$. Lemma 5.4 gives equal finite volume, so the two closed convex bodies agree. Their active facets and reflected sites then agree. $\square$

Theorem 5.6 (exact reconstruction). *Let D be a three-dimensional proper-zero datum. Suppose no vertex of its core passes in $\Gamma(M_J)$. Then the passing vertices of $\Gamma(M_J)$ are exactly the hidden reflected apexes, the core is*

$$P = \bigcap_{i \in I} H_i^- \cap \bigcap_{a \text{ passing}} \{x : \|x - o\| \leq \|x - a\|\},$$

and every compatible decomposition of $X(D)$ is the same. Consequently, $X(D)$ congruent to $X(D')$ implies D congruent to D'.

Proof. Let A be the hidden apex set of D, and let (P'', A'') be any compatible decomposition. The graph $\Gamma(M_J)$ is intrinsic, hence common to both decompositions. Lemma 5.3 gives that every element of A and A'' passes in this common graph. By Lemma 5.2, a passing vertex is an apex of D or a core vertex. The second possibility is excluded by hypothesis, so the passing set is exactly A and $A'' \subseteq A$. Lemma 5.5 completes the uniqueness proof.

An ambient isometry carrying one raw union to another preserves the intrinsic components, centre, exposed sheet germs, and $\Gamma(M_J)$. It carries the first datum to a compatible decomposition of the second raw union, which uniqueness identifies with the second datum. $\square$

6. The exceptional centre set

Theorem 6.1 (fixed-realization genericity). *Fix a realized three-polytope P and a visible-facet mask. The set of centres $o \in \operatorname{int} P$ at which some core vertex passes is contained in a finite union of proper spheres and quadrics. In particular, it has three-dimensional Lebesgue measure zero.*

Proof. Use fixed absolute coordinates. Suppose a core vertex v is incident to a hidden facet with outward unit normal n and equation $n \cdot x = h$. The reflected apex is

$$a = o + 2(h - n \cdot o)n.$$

The two walls meeting along $[v, a]$ have distinct planes, so this is an edge of $\Gamma(M_J)$. If v passes, then $a \in B(o, v)$, which forces

$$4(h - n \cdot o)^2 = \|v - o\|^2.$$

Its quadratic part is $o^\mathsf{T}(4nn^\mathsf{T} - I)o$. The eigenvalue along n is 3 and the two eigenvalues on $n^\perp$ are -1. The polynomial is nonzero and its zero set is a proper quadric.

If v is incident only to visible hinge facets, the core edges at v are intersections of distinct facet planes and are intrinsic crease edges. Any crease neighbour w required by passing satisfies

$$\|w - o\|^2 = \|w - v\|^2,$$

a proper sphere condition in o.

There are finitely many vertices and incidences. Every passing event is contained in at least one of the displayed proper algebraic hypersurfaces, and a finite union of such zero sets has measure zero. □

The theorem deliberately does not assert that the bad-centre set itself is closed. A subset of a closed algebraic union need not be closed. Nor does the argument define a measure on simultaneously varying realizations of P.

7. Ambiguities require five hidden sites per side

This section studies what a failure of uniqueness would have to look like. Throughout, D is a three-dimensional proper-zero datum with decomposition (P, A) and centre o, and (P'', A'') is a second compatible decomposition of the same raw union. By Section 3 the two decompositions share M_J, the centre, the visible hinge facets, and the intrinsic graph $\Gamma(M_J)$.

Lemma 7.1 (differing sites avoid visible hinges)**.** *Let $a \in A'' \setminus A$. Then a is a vertex of P that passes in $\Gamma(M_J)$ and lies on no visible hinge facet. Symmetrically for $A \setminus A''$.*

Proof. As an apex of the second decomposition, a passes (Lemma 5.3) and, by Lemma 5.2 applied to the first decomposition, a is an apex of D or a vertex of P; the first option is excluded by $a \notin A$. Applying Lemma 5.1 in the second decomposition, the only boundary patches containing a are the walls of its own cap there. A visible hinge facet F_i is a boundary patch of every decomposition and is contained in the core of each; since an apex lies strictly outside the core, F_i is not a wall of the cap of a, and hence $a \notin F_i$. □

Theorem 7.2 (hinge-anchored uniqueness)**.** *Suppose every vertex of the core that passes in $\Gamma(M_J)$ lies on at least one visible hinge facet. Then the compatible decomposition is unique. Theorem 5.6 is the special case in which no core vertex passes.*

Proof. Suppose $(P'', A'') \neq (P, A)$. If $A'' \subseteq A$, Lemma 5.5 forces equality, so $A'' \setminus A \neq \varnothing$. Any $a \in A'' \setminus A$ is, by Lemma 7.1, a passing core vertex on no visible hinge facet, contradicting the hypothesis. $\square$

Lemma 7.3 (exchange edges)**.** *Let $a \in A'' \setminus A$. Then a lies on at least three hidden facets of (P, A). For every hidden facet $G_c \ni a$, its site c satisfies $c \in A \setminus A''$, the segment $[a, c]$ is an edge of $\Gamma(M_J)$, and the triangle (o, a, c) is equilateral. Symmetrically with the two decompositions exchanged.*

Proof. By Lemma 7.1, a is a vertex of the three-polytope P, so at least three facets of P pass through it, and none of them is visible; distinct facets have distinct planes, and the site is the reflection of o in the plane, so the corresponding sites are distinct.

Let G_c be such a hidden facet. Then $a \in \operatorname{vert} G_c$, so by the proof of Lemma 5.3 the segment $[c, a]$ is an edge of $\Gamma(M_J)$. The facet plane $\operatorname{aff} G_c$ is the perpendicular bisector of o and c, and a lies on it, so $\|a - o\| = \|a - c\|$. Since a passes and c is one of its graph neighbours, $\|c - o\| = \|c - a\|$. The triangle is equilateral.

Finally $c \notin A''$: otherwise c would be an apex of the second decomposition, and Lemma 5.3 applied there would give $N_{\Gamma(M_J)}(c) = \operatorname{vert} G''_c \subset P''$; but $a \in N_{\Gamma(M_J)}(c)$ is an apex of the second decomposition and lies strictly outside P''. $\square$

Lemma 7.4 (two common neighbours)**.** *Let X be the graph on $(A \setminus A'') \cup (A'' \setminus A)$ whose edges are the pairs $\{a, c\}$ of Lemma 7.3. Then X is simple and bipartite between the two difference sets, with minimum degree at least three, and two distinct sites on one side have at most two common neighbours.*

Proof. Bipartiteness, simplicity and the degree bound are Lemma 7.3. Let c be a common neighbour of the distinct same-side sites a and a'. The two equilateral relations give $\|c-o\| = \|c-a\| = \|a-o\|$ and $\|c-o\| = \|c-a'\| = \|a'-o\|$. So c lies on the perpendicular bisector plane of o and a, on that of o and a', and on the sphere of radius $\|a - o\|$ about o. The two planes are distinct, because reflecting o across them returns a, respectively a'; hence they meet in a line or not at all, and a line meets a sphere in at most two points. $\square$

Lemma 7.5 (incidence count)**.** *Each side of X has at least four sites. If some side has exactly four, then both do, X is 3-regular, that is, $K_{4,4}$ minus a perfect matching, and all eight sites lie at one common distance $R > 0$ from o with every edge a chord of length R.*

Proof. Write p and q for the two side sizes. Every vertex has at least three neighbours, all on the other side, so $p, q \geq 3$. Count the incidences $(\{a, a'\}, c)$ of a common neighbour c with an unordered pair of distinct sites on the p-side: Lemma 7.4 bounds them by $2\binom{p}{2}$, and they number $\sum_c \binom{\deg c}{2} \geq 3q$, because each of the q vertices c of the other side has degree at least three.

If $p = 3$, every vertex of the other side is adjacent to all three sites, so one pair on the p-side has $q \geq 3 > 2$ common neighbours, impossible. Hence, exchanging the roles of the sides, $p, q \geq 4$. If $p = 4$, then $3q \leq 2\binom{4}{2} = 12$ forces $q \leq 4$, hence $q = 4$; and a vertex of degree four on either side would give $\sum_c \binom{\deg c}{2} \geq \binom{4}{2} + 3\binom{3}{2} = 15 > 12$, so every degree is exactly three. The bipartite complement of a 3-regular bipartite graph on $4+4$ vertices is 1-regular, a perfect matching. Such a graph is connected (a component would need at least three vertices on each side), and the equilateral relation of Lemma 7.3 propagates the distance $\|a - o\|$ across every edge, so all eight sites lie at one distance $R > 0$ and every edge has length R. $\square$

The remaining case is a genuinely geometric one, and it is the heart of the bound.

Theorem 7.6 (no exchange quadruple)**.** *There is no configuration of distinct unit vectors $d_1, \ldots, d_4$ and distinct unit vectors $e_1, \ldots, e_4$ in $\mathbb{R}^3$ with*

$$e_l \cdot d_i = \tfrac{1}{2} \qquad \textit{for all } l \neq i.$$

Proof. Write $x_{ij} = d_i \cdot d_j$, let G be the unit-diagonal Gram matrix of $d_1, \ldots, d_4$, let G_i be its principal 3×3 minor deleting row and column i, and let r_i be the deleted off-diagonal column. Define the polynomials

$$\begin{aligned} g_0 &:= \det G, \\ g_i &:= \mathbf{1}^{\mathsf{T}} \operatorname{adj}(G_i)\mathbf{1} - 4 \det G_i \qquad (i = 1, \ldots, 4), \\ s_i &:= \mathbf{1}^{\mathsf{T}} \operatorname{adj}(G_i) r_i - \det G_i. \end{aligned}$$

For example, in the variables (x_{23}, x_{24}, x_{34}),

$$\begin{aligned} g_1 = 3x_{23}^2 + 3x_{24}^2 + 3x_{34}^2 - 8x_{23}x_{24}x_{34} \\ + 2(x_{23}x_{24} + x_{23}x_{34} + x_{24}x_{34}) - 2(x_{23} + x_{24} + x_{34}) - 1. \end{aligned}$$

Step 1: the configuration satisfies the polynomial system. Four vectors in $\mathbb{R}^3$ give $g_0 = \det G = 0$. Fix l and suppose the triple $\{d_i : i \neq l\}$ were linearly dependent; applying $e_l \cdot (\,\cdot\,)$ to a dependence shows the coefficients sum to zero, so three distinct points would be affinely dependent, hence collinear, and a line meets the unit sphere in at most two points. So each triple is a basis and $\det G_l \neq 0$, and e_l is the unique solution of the three linear equations $e_l \cdot d_i = \frac{1}{2}$ $(i \neq l)$. Solving that system by Cramer's rule and imposing $\|e_l\|^2 = 1$ yields $\mathbf{1}^{\mathsf{T}} \operatorname{adj}(G_l)\mathbf{1} = 4 \det G_l$, that is, $g_l = 0$ on the Gram coordinates of the configuration. Moreover $c_l := e_l \cdot d_l - \frac{1}{2}$ satisfies $2c_l \det G_l = s_l$. If some $c_l = 0$, then $e_l \cdot d_i = \frac{1}{2}$ for all four i, so all four d_i lie on the plane $\{x : e_l \cdot x = \frac{1}{2}\}$, which misses the origin; any three of them are then linearly independent, and every e_m solves the same three linear equations as e_l, forcing $e_m = e_l$ for all m and contradicting distinctness. Hence every $s_l \neq 0$.

Step 2: four certificate identities force an isosceles pattern. There exist explicit polynomials $q_0^{(\ell)}, \ldots, q_4^{(\ell)} \in \mathbb{Q}[x_{12}, \ldots, x_{34}]$ of total degree at most eight with

$$\ell \cdot s_1 s_2 \;=\; q_0^{(\ell)} g_0 + q_1^{(\ell)} g_1 + q_2^{(\ell)} g_2 + q_3^{(\ell)} g_3 + q_4^{(\ell)} g_4$$

for each of the four linear forms

$$\ell \in \{\, x_{12} - x_{34},\; x_{13} - x_{24},\; x_{14} - x_{23},\; x_{23} + x_{24} + x_{34} + 1 \,\}.$$

The cofactors are archived, with the verification script, alongside the reference implementation; each identity is re-verified by expanding the difference of the two sides to the zero polynomial in exact rational arithmetic, independently of the software that found it. At the configuration all g_i vanish and $s_1 s_2 \neq 0$, so all four linear forms vanish:

$$x_{12} = x_{34} =: a, \qquad x_{13} = x_{24} =: b, \qquad x_{14} = x_{23} =: c, \qquad a + b + c = -1.$$

Step 3: an elementary emptiness argument. Substituting the pattern into any of the four face polynomials leaves one cubic; explicitly,

$$g_i\big|_{\text{pattern}} = 4F(a,b), \qquad F(a,b) = 2ab^2 + 2a^2 b + a^2 + 3ab + b^2 + a + b + 1,$$

an identity in (a,b) that is checked by direct substitution. Hence $F(a,b) = 0$. By Cauchy–Schwarz $|x_{ij}| \leq 1$, so $(a,b) \in [-1,1]^2$. On the open square F is strictly positive. Indeed the difference $\partial_b F - \partial_a F$ factors as $(a-b)(2a+2b+1)$; on the branch $b = a$ the equation $\partial_a F = 0$ becomes $6a^2+5a+1 = (3a+1)(2a+1) = 0$, and on the branch $2a+2b+1 = 0$ it becomes $-a(2a+1) = 0$. This gives exactly the four interior critical points $(-\frac{1}{3}, -\frac{1}{3}), (-\frac{1}{2}, -\frac{1}{2}), (0, -\frac{1}{2}), (-\frac{1}{2}, 0)$ with values $\frac{20}{27}, \frac{3}{4}, \frac{3}{4}, \frac{3}{4}$, and the edge restrictions of F are

$$F(a,1) = 3(a+1)^2, \qquad F(a,-1) = 1 - a^2,$$
$$F(1,b) = 3(b+1)^2, \qquad F(-1,b) = 1 - b^2,$$

whose only zeros on the closed square are the three corners $(-1,1)$, $(1,-1)$ and $(-1,-1)$ of the (a,b)-square. An interior zero of F would force an interior critical point with a nonpositive value, and there is none. At the three corners, respectively $x_{24} = 1$, $x_{34} = 1$, and $x_{14} = -1 - a - b = 1$, so two of the d_i coincide by the equality case of Cauchy–Schwarz, contradicting distinctness. No configuration exists. □

Remark 7.7. The certificate identities show that the four linear forms, and with them $F(x_{34}, x_{24})$, lie in the saturation of $\langle g_0, \ldots, g_4 \rangle$ by $s_1 s_2$: over the complex numbers, every solution of the distance-$\frac{1}{2}$ system with $s_1 s_2 \neq 0$ is already isosceles-patterned. The certificates were found by a Gröbner-basis saturation in Singular 4.3.2, with the saturation recomputed by iterated ideal quotients under in-run containment and stability checks, and the computed reduced basis of the fully saturated ideal is $\{x_{23} + x_{24} + x_{34} + 1,\; x_{14} + x_{24} + x_{34} + 1,\; x_{13} - x_{24},\; x_{12} - x_{34},\; F(x_{34}, x_{24})\}$; the proof above uses only the certified fragment, re-verified by exact expansion in SymPy 1.14, so no Gröbner implementation is trusted. As a standing positive control, the same pipeline at the distance $\frac{1}{3}$ keeps the regular tetrahedron on the saturated variety, as it must.

Theorem 7.8 (five sites per side)**.** *Two distinct compatible decompositions of one proper-zero raw union differ in at least five reflected sites on each side:* $|A \setminus A''| \geq 5$ *and* $|A'' \setminus A| \geq 5$.

Proof. As in Theorem 7.2, distinctness makes both difference sets nonempty. By Lemma 7.5 each side of the exchange graph X has at least four sites, and if some side had exactly four, X would be $K_{4,4}$ minus a perfect matching with all eight sites at one distance $R > 0$ from o and all twelve edges of length R. Translating o to the origin and scaling by R^{-1} makes the sites unit vectors with $\|a - c\| = 1$, i.e. $a \cdot c = \frac{1}{2}$, along every edge; labelling the matching pairs with equal indices produces exactly the configuration of Theorem 7.6, which does not exist. Hence both sides have at least five sites. □

Corollary 7.9 (at most four hidden facets)**.** *A proper-zero datum with at most four hidden facets has a unique compatible decomposition, with no hypothesis on its centre.*

Proof. A second decomposition would give $|A \setminus A''| \geq 5$, while $|A| \leq 4$. □

8. Exact reconstruction algorithm and evidence

Assume an exact rational polygonal presentation of the exposed boundary. Coplanar subdivisions are presentation data and may be inserted or removed. The conceptual algorithm is:

1. Compute the closures of the connected components of the local-flat locus. With at least three components, select the unique centre of their codimension-one adjacency star; with exactly two, select the component of larger intrinsic volume. This is M_J.
2. Intersect every other component with M_J to recover the nonflat hinges and their cap geometry.
3. Recover o from either nonflat cap by the two-candidate construction in Proposition 3.1.
4. Construct $\Gamma(M_J)$ from exposed sheet germs.
5. Keep exactly the graph vertices that pass.
6. Intersect the visible hinge halfspaces with the bisector halfspaces of the passing vertices, the Dirichlet-cell construction of [3].

Under the hypothesis of Theorem 5.6, correctness is immediate from Lemmas 5.2 and 5.3. A full certifier may rebuild the entire raw union, including the observed nonflat caps, and compare it exactly with the input. The reference code implements the flat reconstruction kernel in steps 4–6; it is not an end-to-end implementation of steps 1–3 or of the full raw-union certifier.

The certifying variant compares two intrinsic conditions, and each is provably necessary. There are constructed exact data, at exceptional centres, on which the reconstruction returns a strict superset of the true site set while the rebuilt merged flat component still equals the observed one exactly:

cutting a passing vertex off with its bisector plane and re-attaching the cap over the new facet restores the removed piece. Comparing only the merged flat component therefore certifies a wrong answer on those data (six exact witnesses). What the wrong reconstruction destroys is the recovered visible hinge facets, so the certifier also requires every recovered visible hinge facet to survive as a facet of the reconstructed core. On a second witness family (three exact witnesses) the situation is reversed: the hinge facets survive while the flat components differ. Hence neither condition alone suffices, and the certifier checks both.

Let N be the number of input boundary corners and k the number of returned core facets. The reference germ extractor atomises each segment against all exact endpoints and builds carrier labels during atomisation. Its conservative bound is $O(N^2 \log N)$ exact field operations and $O(N^2)$ crease atoms. The reference three-dimensional halfspace kernel enumerates the $\binom{k}{3}$ plane triples and validates each candidate corner against every halfspace, so it costs $O(k^4)$. Thus the proved reference bound is

$$O(N^2 \log N + k^4).$$

No sharper literature bound is imported into this statement.

The executable audit has two layers. E-A60 runs the intrinsic germ skeleton on every frozen six-vertex row. E-A58 runs the reconstruction/flat certification kernel. The frozen measurements are:

intrinsic-skeleton corpus rows	16,744
rows reconstructed by the intrinsic signature	16,744
rows agreeing with the historical graph on the corpus	16,744
subdivision mutations invariant (256-row corpus prefix)	256/256
maximal-sheet cross-checks (same prefix)	256/256
flat-kernel reconstructions	16,744/16,744
sampled two-condition certifications	340/340
wrong answers certified	0
degenerate swallowed-corner witness	pass
exceptional-centre witnesses	6
flat-only accepts / two-condition rejects	6/6
reversed witnesses (hinges survive, flat differs)	3/3
corpus disagreements between the two certifier variants	0/340
certificate identities of Theorem 7.6 verified	4/4

The canonical payloads are, for the intrinsic-skeleton audit E-A60,

`D18C07176BAF7B729A550E9B9B8B9D9E04BB58A0C52D295243108417C9BE785A`,

for the two-condition reconstruction/certification kernel E-A58,

`3C58C0B88081380C7D22993A6FEE3B4EC9FE1A8C3D0CDC107BE9D40C33BA4ACC`,

for the exceptional-centre adversarial control E-A61,

`76956805B0BA304142CB08D833FA1179876A9F2154A76CE6C5A232BEF9E3F7F1`,

for the reversed witness family E-A63,

`50B26633575C81C67AD3A9857DC5BE1227B512E21CC98168EECEC37B360DB1A8`,

and for the certificate verification of Theorem 7.6, whose payload pins the SHA-256 of all four cofactor files, E-A69,

5DC4C5800572C64F07C7A815CE7BEFFB930FEF03E04CC71679FB9501606F1D92.

Finite replay is evidence about the implementation and its frozen domain. The proof of Theorem 5.6 is the intrinsic argument in the preceding sections, and the proof of Theorem 7.6 rests on the verified identities and the elementary analysis of Section 7, not on the replay.

9. Relation to supplied-boundary inverse problems

Ash and Bolker study recognition of supplied Dirichlet tessellations. Their Lemma 1 places a common cell boundary on the perpendicular bisector of its two sources, and their Theorem 2 constructs a Dirichlet cell by reflecting an interior point in the known facet hyperplanes [3]. The latter is exactly the forward reflection construction used in Section 2 and is cited as such.

Subsequent recognition algorithms also begin with substantially visible boundary data. Aurenhammer takes a supplied polytopical cell complex [4]; Hartvigsen takes a tessellation of Euclidean space into polyhedra and uses linear programming [15]; and Schoenberg, Ferguson, and Li recover generators from segments demarcating cell boundaries [18]. In the plane, Alonso Ferrero states the generator-recognition problem with the Voronoi edges provided [13]. Biedl, Held, and Huber reconstruct from a supplied embedded straight-line graph with rays [5]. Power-diagram detection and weighted bisector-graph reconstruction similarly start from a supplied complex, partition, or geometric graph [7, 12]. The generalized inverse Voronoi problem studied by Aloupis et al. permits added edges while retaining the original input edges [2].

Several further inverse-tessellation methods reinforce this distinction. Suzuki and Iri approximate a supplied planar tessellation by a Voronoi diagram, with exact generator restoration included as a special case [19]. Duan et al. recover weighted generators from a supplied Laguerre tessellation and show that those generators are generally nonunique [11]. Chaidee and Sugihara recognise spherical Laguerre Voronoi diagrams from a supplied spherical tessellation [10]. Most recently, Hernandez-Suarez reconstructs the generators of a supplied planar Voronoi tessellation from one interior cell and its neighbours, then propagates them by successive reflections [16]. In every case the diagram, partition, or tessellation boundaries remain part of the input.

Not every inverse tessellation problem receives boundary edges. Bourne, Pearce, and Roper prove that labelled Laguerre cells are uniquely determined by their volumes and centroids and recover them by convex optimization [9]. Bourne, Mulholland, Sahu, and Tant instead fit an oriented planar Voronoi diagram from ultrasonic boundary travel times in a simplified nondestructive-testing model; the geometry is unknown, their first objective can have nonunique minimizers, and the reported reconstruction is noise-sensitive [8]. These are partial- and indirect-data comparators, but their observables are respectively global labelled cell moments and travel times, not a reflected-cap raw union.

Three further classical theories are sometimes suggested as antecedents. They concern different observables: the theory of convex polyhedra from intrinsic surface data [1], the reconstruction of a simple polytope from its graph [6, 17], and geometric tomography, which works from lower-dimensional data such as projections and sections [14]. None of these takes the reflected-cap raw union as its input, and no comparison beyond that input-model distinction is made here.

The precise distinction in the present model is that the zero-angle source layer and its facet boundaries have already been erased by the raw union. The intrinsic germ skeleton is designed to use only the boundary information that survives that erasure. This comparison is a statement about input hypotheses, not a claim that the present result is the first possible treatment of every related erased-boundary problem.

10. Limitations and open problems

The following questions are not resolved here.

- Whether an exact ambiguity exists at all remains open. Theorems 7.2 and 7.8 constrain it: every ambiguous datum needs a passing core vertex avoiding all visible hinge facets and at least five differing sites on each side, hence at least five hidden facets. Data with five or more hidden facets are not decided here.
- The proof is three-dimensional. Higher-dimensional analogues require a new analysis of the boundary-germ strata and candidate types.
- Theorem 7.6 is computer-assisted: the four certificate identities were found by machine and are re-verified by exact expansion. The verification chain ships with the reference implementation; the remaining steps are hand-checkable.
- The reference implementation assumes an exact polygonal presentation. It does not extract such a presentation from a point cloud, floating-point mesh, or arbitrary implicit-set oracle.
- E-A58 implements the flat reconstruction kernel and the two-condition certifier; it is not the full six-step pipeline or a raw-union certifier. Completeness of the two conditions is not claimed; each is necessary, and their conjunction rejects every constructed wrong reconstruction.
- The finite corpus and deterministic mutation sample do not establish novelty, priority, or unrestricted-domain behaviour.

References

[1] A. D. Alexandrov. *Convex Polyhedra*. Springer Monographs in Mathematics. Springer, 2005.

[2] Greg Aloupis, Hebert Pérez-Rosés, Guillermo Pineda-Villavicencio, Perouz Taslakian, and Dannier Trinchet-Almaguer. Fitting voronoi diagrams to planar tessellations. *arXiv preprint arXiv:1308.5550*, 2013.

[3] Peter F. Ash and Ethan D. Bolker. Recognizing dirichlet tessellations. *Geometriae Dedicata*, 19:175–206, 1985.
[4] Franz Aurenhammer. Recognising polytopical cell complexes and constructing projection polyhedra. *Journal of Symbolic Computation*, 3:249–255, 1987.
[5] Therese Biedl, Martin Held, and Stefan Huber. Recognizing straight skeletons and voronoi diagrams and reconstructing their input. In *10th International Symposium on Voronoi Diagrams in Science and Engineering*, pages 37–46, 2013.
[6] Roswitha Blind and P. Mani-Levitska. Puzzles and polytope isomorphisms. *Aequationes Mathematicae*, 34:287–297, 1987.
[7] Steffen Borgwardt and Rafael M. Frongillo. Power diagram detection with applications to information elicitation. *arXiv preprint arXiv:1711.06207*, 2017.
[8] David P. Bourne, Anthony J. Mulholland, Smita Sahu, and Katherine M. M. Tant. An inverse problem for voronoi diagrams: A simplified model of non-destructive testing with ultrasonic arrays. *Mathematical Methods in the Applied Sciences*, 44(5):3727–3745, 2021.
[9] David P. Bourne, Mason Pearce, and Steven M. Roper. Inverting laguerre tessellations: Recovering tessellations from the volumes and centroids of their cells using optimal transport. *ESAIM: Mathematical Modelling and Numerical Analysis*, 59(2):841–871, 2025.
[10] Supanut Chaidee and Kokichi Sugihara. Recognition of the spherical Laguerre Voronoi diagram. *arXiv preprint arXiv:1705.03911*, 2017.
[11] Qibin Duan, Dirk P. Kroese, Tim Brereton, Aaron Spettl, and Volker Schmidt. Inverting Laguerre tessellations. *The Computer Journal*, 57(9):1431–1440, 2014.
[12] Günther Eder, Martin Held, Stefan de Lorenzo, and Peter Palfrader. On the recognition and reconstruction of weighted voronoi diagrams and bisector graphs. *Computational Geometry*, 109:101935, 2023.
[13] M. Montserrat Alonso Ferrero. Voronoi diagram: The generator recognition problem. *arXiv preprint arXiv:1105.4246*, 2011.
[14] Richard J. Gardner. *Geometric Tomography*. Cambridge University Press, 2 edition, 2006.
[15] David Hartvigsen. Recognizing voronoi diagrams with linear programming. *ORSA Journal on Computing*, 4(4):369–374, 1992.
[16] Carlos M. Hernandez-Suarez. Fast and accurate reconstruction of Voronoi generators in large tessellations. *arXiv preprint arXiv:2506.19076*, 2025.
[17] Gil Kalai. A simple way to tell a simple polytope from its graph. *Journal of Combinatorial Theory, Series A*, 49(2):381–383, 1988.
[18] Frederic Paik Schoenberg, Thomas Ferguson, and Cheng Li. Inverting dirichlet tessellations. *The Computer Journal*, 46(1):76–83, 2003.
[19] Atsuo Suzuki and Masao Iri. Approximation of a tessellation of the plane by a Voronoi diagram. *Journal of the Operations Research Society of Japan*, 29(1):69–97, 1986.

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